

Traversability Estimation

Our RGB was used to do Perception related activities like Detection, Segmentation and localization as well . But we wanted to add a brain to it .

It was not intelligent enough to estimate if the zone infront is traversable or Non - Traversable.

So We started with the Go-Net model to implement it for our usecase.

This project is mostly a Data Engineering and Model Training project where i took an old model described in their paper and tried to train it for our own usecase (warehouse).

Go-Net model : <https://cvgl.stanford.edu/gonet/>

Biggest hurdle in this project was the dataset that they used has different modality that is fisheye camera and we here are using pin hole camera. But i had to leverage their huge dataset to make the model generalizable - still working on it.

Thinking of implementing this : <https://sites.google.com/view/exaug-nav> to augment the data !

Here is a small snippet of that work :

Thumbnail :

